DILI — Valproate Mitochondrial Hepatotoxicity

1. Valproate bottle pouring

WHAT YOU SEE

VALPROATE bottle pouring into the giant mitochondrion

WHAT IT ENCODES

Valproic acid (VPA) is an antiepileptic/mood stabilizer that causes idiosyncratic, dose-independent hepatotoxicity. The toxic 4-ene-VPA metabolite (formed via CYP) is conjugated to and depletes carnitine and coenzyme A, which crippling mitochondrial beta-oxidation of fatty acids. With fat unable to be oxidized, free fatty acids back up inside hepatocytes as tiny droplets -> microvesicular steatosis. Highest risk is in children <2 years on polytherapy.

BOARD TRAP

WRONG: 'Valproate hepatotoxicity is predictable from a high serum drug level, so check a VPA trough.' Serious VPA hepatotoxicity is idiosyncratic and NOT dose/level dependent — a therapeutic level does not exclude it. Discriminator: the diagnosis is clinical (rising AST/ALT, jaundice, malaise, ammonia), not a drug level.

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2. Giant mitochondrion factory

WHAT YOU SEE

Huge orange mitochondrion as the factory floor

WHAT IT ENCODES

The lesion is fundamentally MITOCHONDRIAL: VPA inhibits fatty-acid beta-oxidation and the urea cycle (inhibits carbamoyl phosphate synthetase I), depleting carnitine. Result is microvesicular fat plus hyperammonemia. This is why VPA toxicity resembles Reye syndrome and acute fatty liver of pregnancy, which are also mitochondrial beta-oxidation failures.

BOARD TRAP

WRONG: 'Hepatocellular VPA injury is from a hypersensitivity/allergic mechanism, so give corticosteroids.' The mechanism is metabolic/mitochondrial, not immunoallergic — steroids are not the treatment. Discriminator: no eosinophilia/rash/fever triad; instead you see hyperammonemia and microvesicular steatosis. Treatment is stopping VPA + L-carnitine.

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3. VALPROATE = TOXIN chalkboard

WHAT YOU SEE

Chalkboard: VALPROATE = TOXIN (toxin underlined in red)

WHAT IT ENCODES

VPA behaves as a metabolic mitochondrial toxin. Asymptomatic transaminase elevation is common (up to ~40%), but clinically significant hepatotoxicity is rare (~1/20,000 overall, far higher in young children on polypharmacy) and can be fatal. Always counsel patients on symptoms of liver failure (lethargy, vomiting, jaundice, encephalopathy).

BOARD TRAP

WRONG: 'A mild rise in transaminases on VPA always mandates immediate discontinuation.' Most mild, asymptomatic elevations are transient and adaptive. Discriminator: stop only for symptomatic hepatitis, jaundice, coagulopathy, or enzymes >=3-5x ULN — not for an isolated minor bump.

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4. Microvesicular droplets

WHAT YOU SEE

Round structure filled with tiny fat droplets, MICROVESICULAR

WHAT IT ENCODES

Hallmark histology = MICROVESICULAR steatosis: many small lipid droplets that keep the nucleus CENTRAL (vs macrovesicular = single large droplet pushing nucleus to the periphery). The microvesicular family = VPA, tetracycline, Reye syndrome, NRTIs (didanosine/stavudine), and acute fatty liver of pregnancy — all mitochondrial beta-oxidation poisons.

BOARD TRAP

WRONG: 'Biopsy shows a single large fat vacuole displacing the nucleus, consistent with valproate.' That is MACROvesicular steatosis, which points to alcohol, amiodarone, methotrexate, or NAFLD. Discriminator: VPA = MICROvesicular (tiny droplets, central nucleus).

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5. 5-90 days calendar

WHAT YOU SEE

Calendar: 5–90 DAYS latency with confused man

WHAT IT ENCODES

Idiosyncratic VPA hepatotoxicity has a latency typically within the first few weeks up to ~3 months (most within first 6 months) of starting therapy. The confused man depicts VPA-induced hyperammonemic encephalopathy, which can occur even with normal LFTs and normal VPA levels.

BOARD TRAP

WRONG: 'Confusion on valproate with a normal ammonia rules out a VPA cause.' VPA can cause encephalopathy via mechanisms beyond ammonia, and ammonia should be specifically ordered. Discriminator: in a confused patient on VPA, check an ammonia level even if LFTs are normal — VPA hyperammonemic encephalopathy is treated with VPA cessation + L-carnitine.

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6. APAP direct-toxic comparison

WHAT YOU SEE

Lab tech with APAP and DIRECT TOXIC label

WHAT IT ENCODES

Contrast in DILI mechanism: acetaminophen (APAP) = DIRECT/intrinsic, dose-dependent, predictable toxin (NAPQI depletes glutathione, zone 3 centrilobular necrosis), with antidote N-acetylcysteine. VPA = IDIOSYNCRATIC, host-dependent. Knowing which bucket a drug falls in drives both prediction and antidote choice.

BOARD TRAP

WRONG: 'Give N-acetylcysteine for valproate hepatotoxicity.' NAC is the antidote for acetaminophen, not VPA. Discriminator: the VPA-specific adjunct is L-carnitine; NAC may be used in acute liver failure generally but is not VPA-specific.

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7. Idiosyncratic worker

WHAT YOU SEE

Slumped figure labeled IDIOSYNCRATIC, INH bottles

WHAT IT ENCODES

VPA hepatotoxicity is largely idiosyncratic/host-dependent (genetics, age, polypharmacy), not predictable from dose alone. Isoniazid (INH) shown alongside is the classic OTHER idiosyncratic hepatotoxin (also age-dependent risk, hepatocellular pattern). Both contrast with the direct/dose-dependent APAP.

BOARD TRAP

WRONG: 'Routine scheduled LFT monitoring reliably prevents fatal idiosyncratic DILI from drugs like VPA/INH.' Idiosyncratic injury can progress faster than scheduled labs catch it; symptom-based education is essential. Discriminator: teach patients warning symptoms; do not rely on periodic LFTs alone to prevent fulminant injury.

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8. 45% adaptive gauge

WHAT YOU SEE

Green gauge ~45% ADAPTIVE

WHAT IT ENCODES

Asymptomatic, transient aminotransferase elevation occurs in roughly a third to 40% of VPA users and usually 'adapts'/resolves with continued therapy or dose reduction. This benign adaptation must be distinguished from true progressive hepatotoxicity (rising bilirubin/INR, symptoms).

BOARD TRAP

WRONG: 'Any transaminase elevation on VPA is dangerous adaptation failure requiring drug stoppage.' Adaptation (down-trending, asymptomatic, normal synthetic function) is benign. Discriminator: synthetic dysfunction (high INR, high bilirubin, hypoglycemia, encephalopathy) signals true hepatotoxicity, not benign adaptation.

9. POLG1 scroll

WHAT YOU SEE

Scroll labeled POLG1

WHAT IT ENCODES

POLG encodes mitochondrial DNA polymerase gamma. POLG mutations (e.g., Alpers-Huttenlocher syndrome) dramatically increase the risk of fatal VPA hepatotoxicity and are a CONTRAINDICATION. POLG testing is recommended before VPA in children with suspected mitochondrial disease or unexplained encephalopathy/seizures.

BOARD TRAP

WRONG: 'Valproate is the antiepileptic of choice for a toddler with refractory seizures and developmental regression.' That picture suggests a POLG/mitochondrial disorder, in which VPA can precipitate fatal liver failure. Discriminator: in suspected mitochondrial disease, AVOID valproate and test POLG first.

10. HLA-DR4 goblin

WHAT YOU SEE

Green goblin with HLA-DR4 tag

WHAT IT ENCODES

Host genetic susceptibility (specific HLA alleles) contributes to idiosyncratic DILI risk for many drugs (e.g., HLA-B*57:01/abacavir, HLA-B*15:02/carbamazepine). The goblin reinforces that idiosyncratic injury is driven by host immunogenetics, not just the drug.

BOARD TRAP

WRONG: 'Idiosyncratic DILI is purely random and has no genetic predictors.' Specific HLA haplotypes meaningfully predict risk for several drugs. Discriminator: 'idiosyncratic' means unpredictable from DOSE, not free of genetic/host determinants.

11. 4-pentenoic acid

WHAT YOU SEE

Sign 4-PENTENOIC ACID near the goblin

WHAT IT ENCODES

The hepatotoxic VPA metabolites 4-ene-VPA and 4-pentenoic acid (formed via CYP-mediated desaturation) directly inhibit mitochondrial beta-oxidation enzymes and sequester CoA/carnitine. Enzyme-inducing co-medications (phenytoin, phenobarbital, carbamazepine) increase formation of these toxic metabolites, explaining higher risk on polytherapy.

BOARD TRAP

WRONG: 'Adding an enzyme-inducing antiepileptic lowers valproate toxicity by speeding its clearance.' Inducers shunt VPA toward the toxic 4-ene metabolite pathway, INCREASING hepatotoxicity risk. Discriminator: polytherapy with inducers is a major risk factor for fatal VPA liver injury.

12. ALT/ALP cholestatic dial

WHAT YOU SEE

Alarm: ALT vs ALP, <2 CHOLESTATIC / >5 HEPATOCELLULAR dial

WHAT IT ENCODES

R-value = (ALT/ULN) / (ALP/ULN): R>=5 = HEPATOCELLULAR (VPA's pattern), R<=2 = CHOLESTATIC, 2-5 = mixed. VPA produces a hepatocellular (ALT-predominant) injury, so it sits at the high end of this dial.

BOARD TRAP

WRONG: 'Valproate causes a cholestatic injury with high ALP and bland transaminases.' VPA is hepatocellular (high ALT, R>=5). Discriminator: cholestatic/mixed DILI points to drugs like amoxicillin-clavulanate, erythromycin, or anabolic steroids — not VPA.

13. 5x ULN red alarm

WHAT YOU SEE

Red wall alarm box: 5x ULN

WHAT IT ENCODES

STOP valproate immediately for symptomatic hepatitis, jaundice, rising bilirubin/INR, hypoglycemia, encephalopathy, or transaminases >=5x ULN (>=3x ULN if symptomatic, per Hy's Law concern). Hy's Law (ALT>3x ULN + bilirubin >2x ULN without obstruction) predicts >=10% mortality.

BOARD TRAP

WRONG: 'Continue valproate through jaundice as long as you trend labs weekly.' Jaundice + transaminitis (Hy's Law) signals high mortality and mandates immediate cessation. Discriminator: the threshold to stop is much lower when symptoms/jaundice/coagulopathy are present than for an asymptomatic enzyme bump.

14. IV carnitine truck

WHAT YOU SEE

Delivery truck labeled C / IV CARNITINE

WHAT IT ENCODES

Management: STOP VPA, supportive care, and IV L-CARNITINE (levocarnitine ~100 mg/kg load then ~15 mg/kg q4-6h, or ~50-100 mg/kg/day). Carnitine is depleted by VPA and its repletion restores beta-oxidation and accelerates ammonia clearance; it is most beneficial in severe hepatotoxicity and VPA-induced hyperammonemic encephalopathy.

BOARD TRAP

WRONG: 'The antidote for valproate hepatotoxicity/hyperammonemia is N-acetylcysteine.' The VPA-specific therapy is L-CARNITINE (plus stopping the drug). Discriminator: NAC = acetaminophen; L-carnitine = valproate. Mixing these up is the classic board trap.